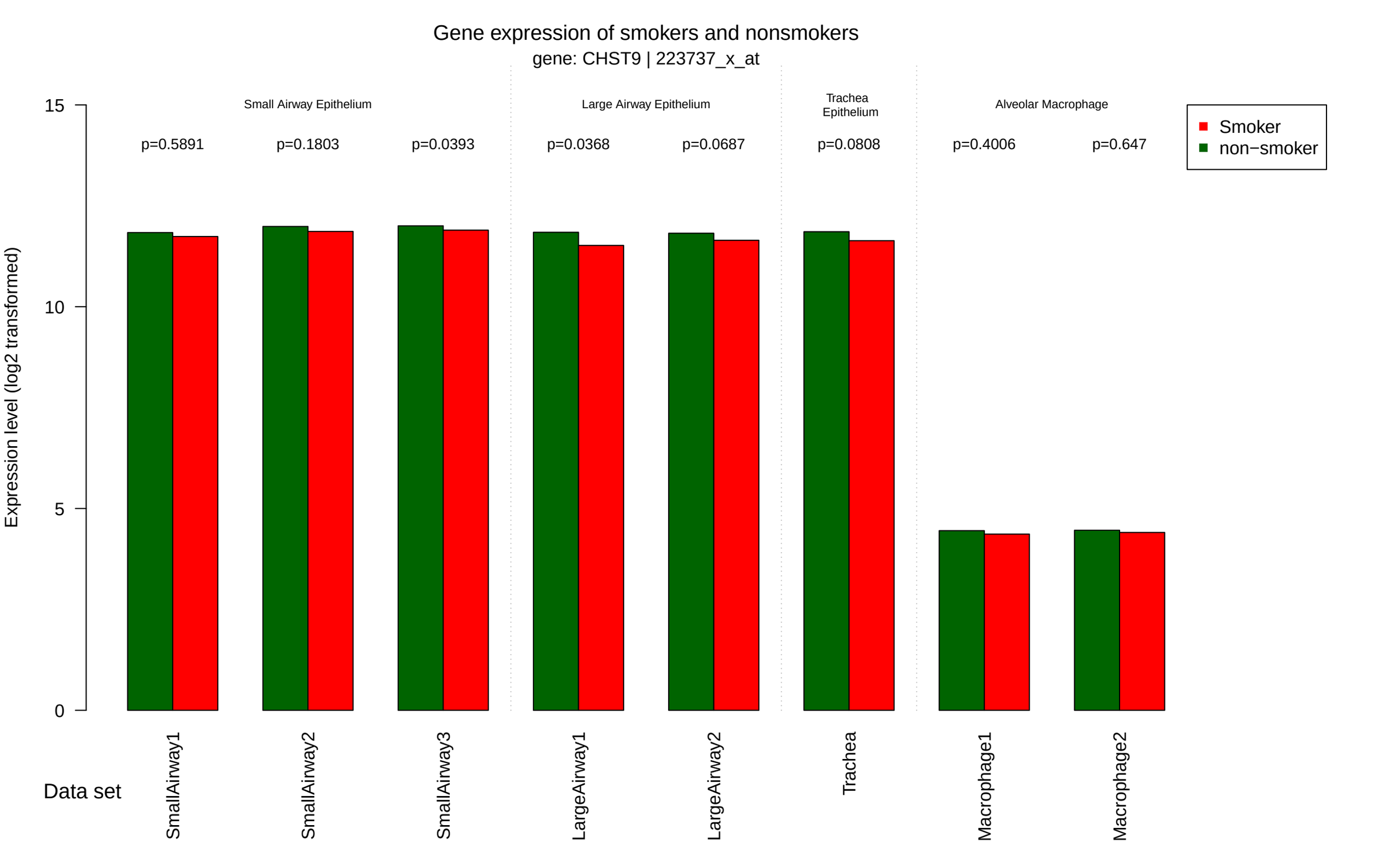
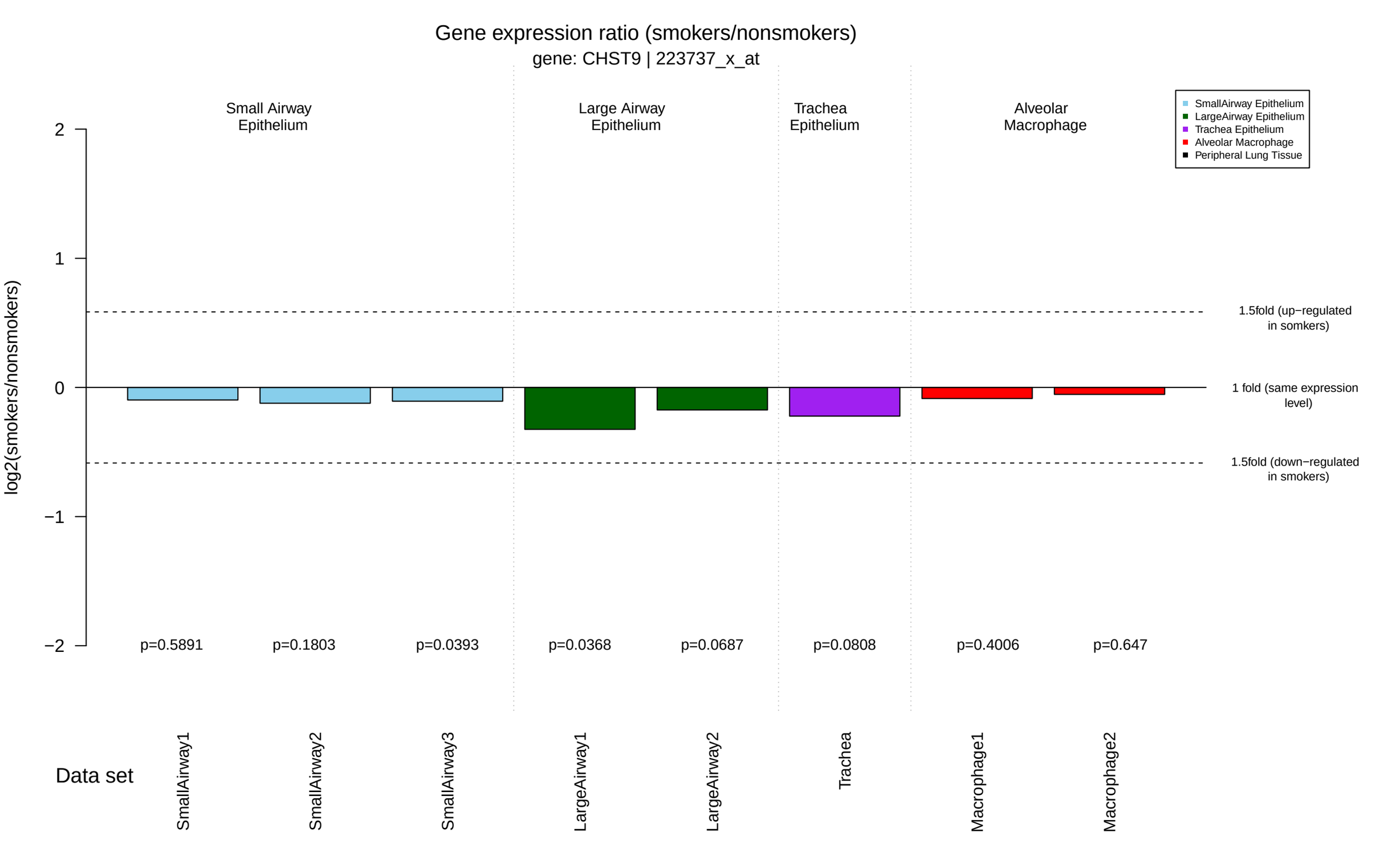
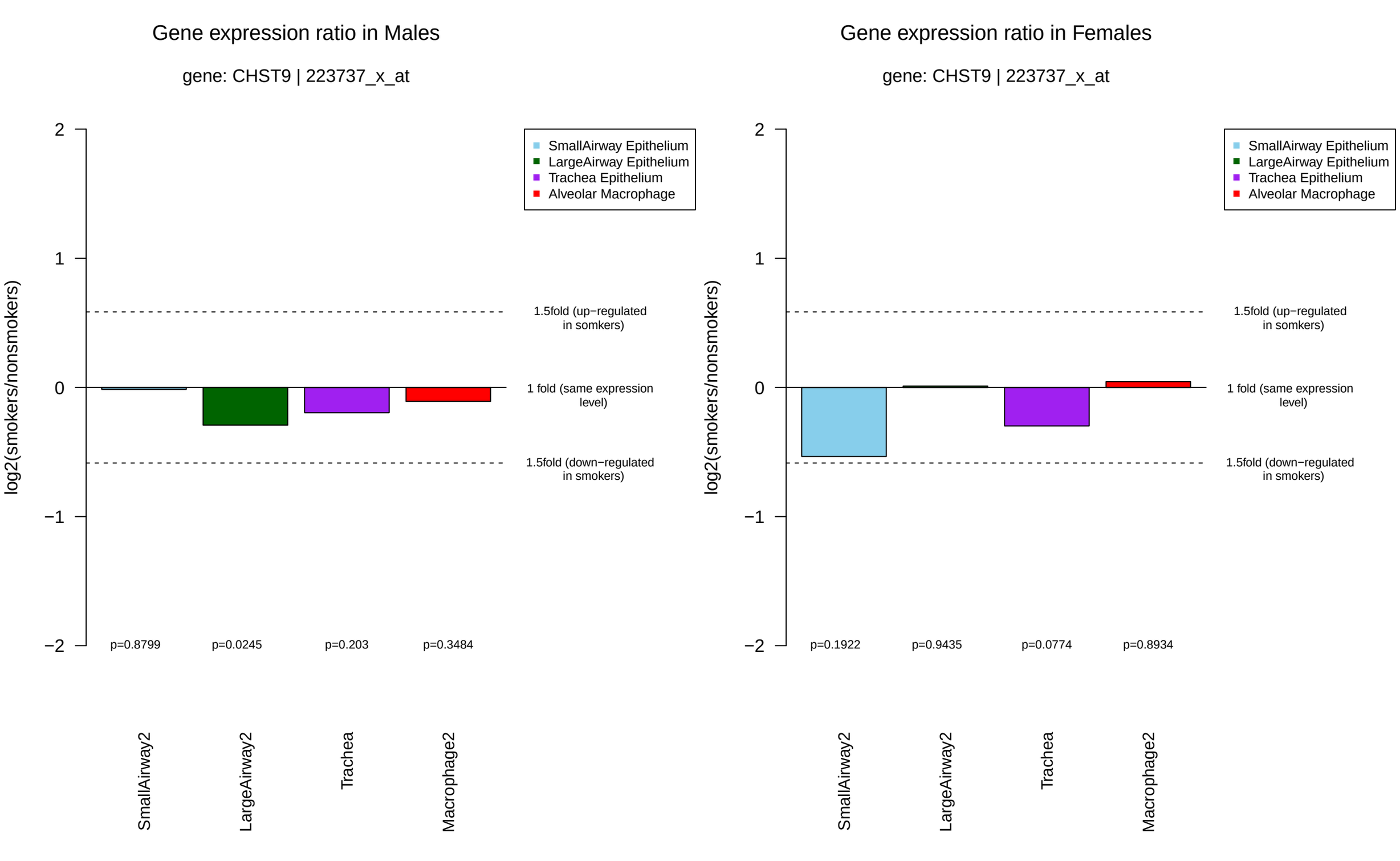
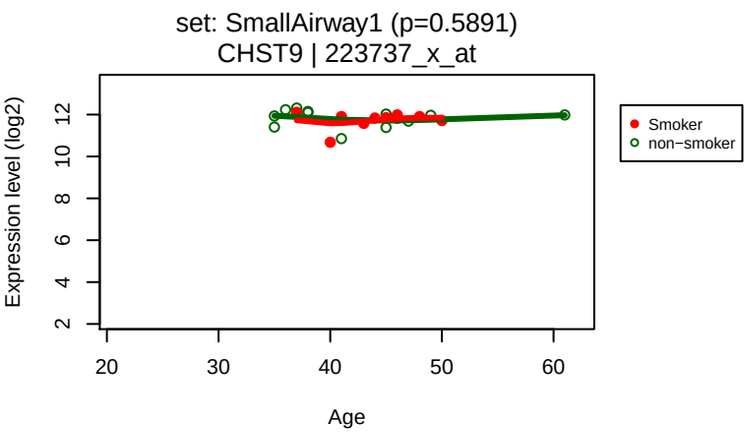
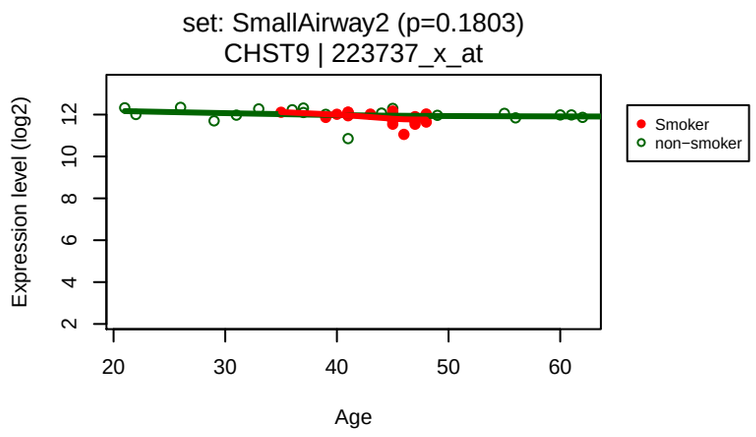
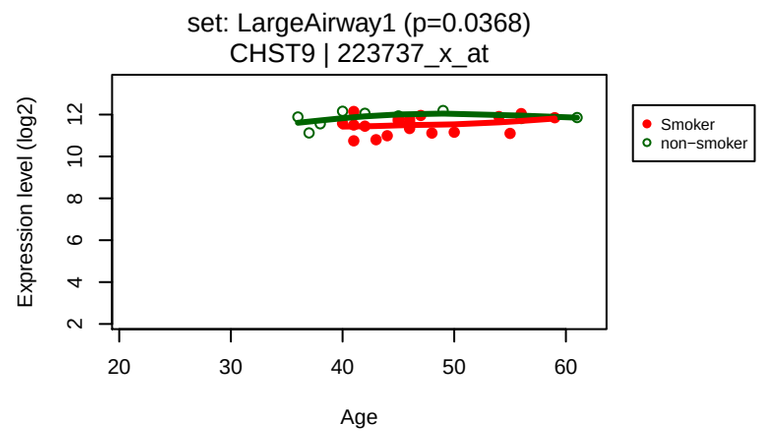

Gene expression of smokers and nonsmokers
gene: CHST9 | 223737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5891
p=0.1803
p=0.0393
p=0.0368
p=0.0687
p=0.0808
p=0.4006
p=0.647
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

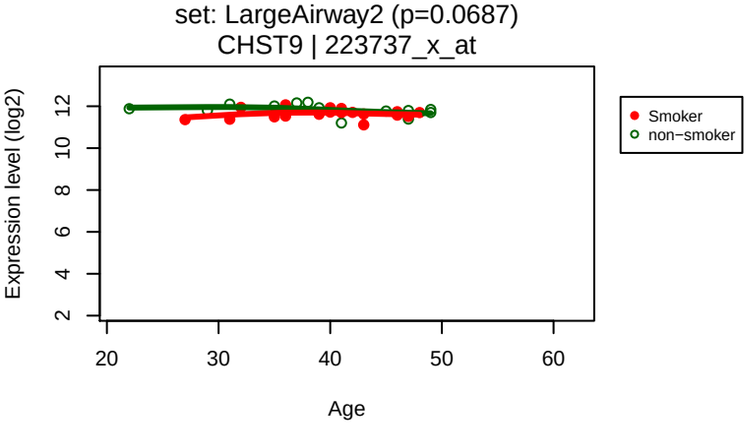

Gene expression ratio (smokers/nonsmokers)
gene: CHST9 | 223737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5891
p=0.1803
p=0.0393
p=0.0368
p=0.0687
p=0.0808
p=0.4006
p=0.647
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

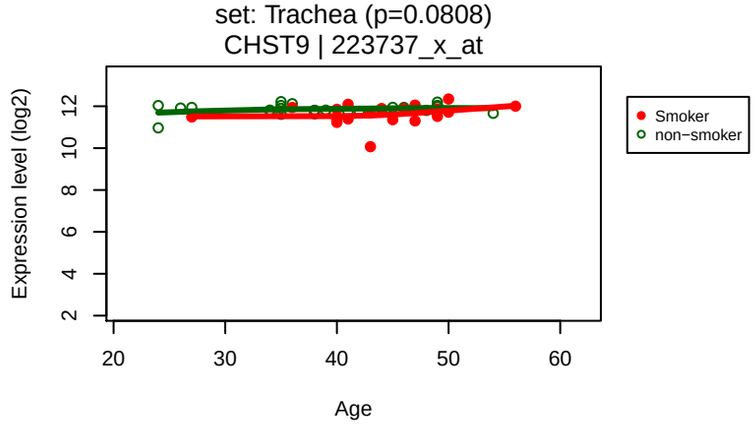

Gene expression ratio in Males
gene: CHST9 | 223737_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8799
p=0.0245
p=0.203
p=0.3484
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST9 | 223737_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1922
p=0.9435
p=0.0774
p=0.8934
SmallAirway2
LargeAirway2
Trachea
Macrophage2

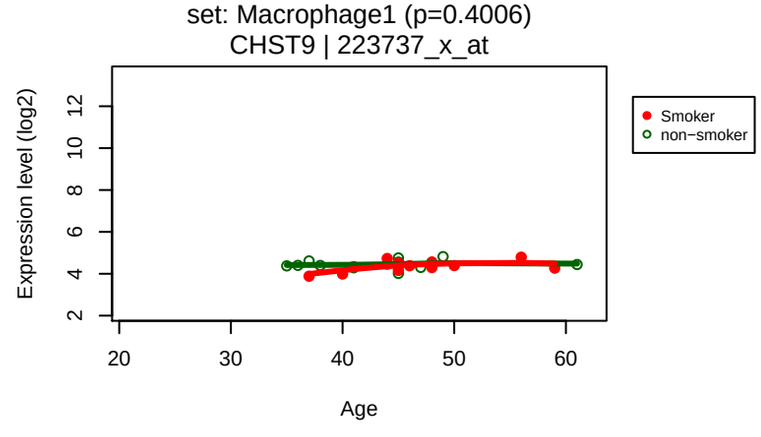

set: SmallAirway1 (p=0.5891)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker

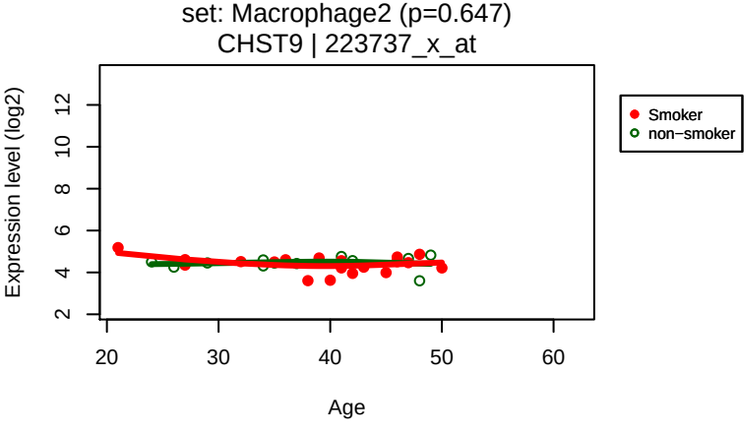

set: SmallAirway2 (p=0.1803)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker

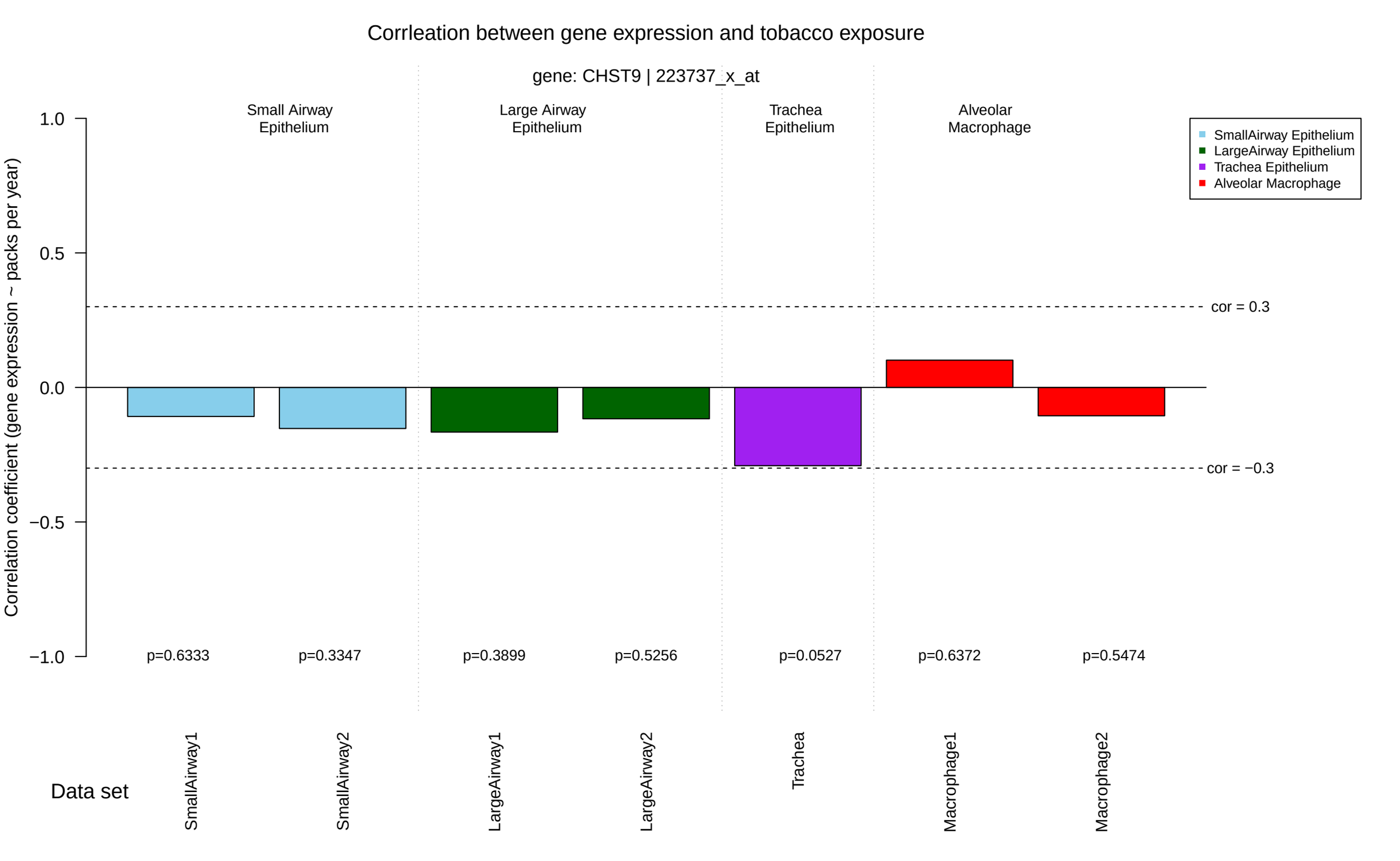

set: LargeAirway1 (p=0.0368)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0687)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0808)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4006)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.647)
CHST9 | 223737_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST9 | 223737_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6333
p=0.3347
p=0.3899
p=0.5256
p=0.0527
p=0.6372
p=0.5474
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2